Two-loop Type II amplitudes for $gg \to AA$ near threshold and the double parton scattering singularity

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In the limit of a heavy top quark, the two-loop amplitudes for $gg \to AA$ contain a class of diagrams, Type II, in which each pseudoscalar couples to the gluons through its own effective vertex. Their finite remainders are known in closed form, but their behaviour near the pair production threshold has not been studied. I expand them in the pseudoscalar velocity β_A at fixed scattering angle through β_A^6. Once the endpoint constants are expressed in a basis of eight transcendental constants, all inverse powers of β_A and the odd powers through β_A^5 cancel identically. At leading power the angle enters only through the logarithm of the transverse momentum p_T of the pseudoscalars. I identify this logarithm with the double parton scattering singularity of one-loop amplitudes: in the forward limit, and with the overall constant fixed at threshold, the velocity dependence of its one-loop coefficient follows the factorised formula of Gaunt and Stirling, including the helicity weights of gluon splitting and the sign structure of a CP-odd coupling, to 3×10^{-15}. An explicit one-loop calculation of $gg \to HH$ through two CP-even vertices gives the corresponding scalar weight with the same constant. At two loops the equal helicity amplitude grows as $\ln^3 p_T$, and the opposite helicity amplitude, which vanishes as β_A^2 at tree level and has no logarithm at one loop, as $\ln^2 p_T$. In the finite virtual corrections to the cross section the logarithm cancels at one loop; at two loops $\ln^3 p_T$ cancels and the highest power is $\ln^2 p_T$, with coefficient $-\frac{7}{4}\pi^2 N_c^2$. I also find that the published two-loop remainders are larger by a factor of 64 than the lower orders and that the published one-loop remainders omit their dependence on the renormalisation scale; all results are given in a consistent normalisation. The series reproduce the unexpanded remainders to 6.77×10^{-5} or better for $\beta_A \leq 0.2$ and are released with an evaluator.

I. INTRODUCTION

Neutral Higgs boson pairs of a two-Higgs-doublet model are produced at the LHC mainly by gluon fusion through top quark loops [1, 2]. For pairs of pseudoscalars, $gg \to AA$, the next-to-leading order (NLO) QCD corrections were first computed in the limit of a heavy top quark [3], later matched to parton showers [4], and recently obtained with the full dependence on the top quark mass [2]. In the benchmark point of Ref. [2] the full NLO cross section for AA lies about 5% below the Born-improved heavy top quark approximation at LHC energies, against about 12% for hH, so the heavy top quark limit, with the exact mass dependence kept at leading order, is a reasonable starting point for higher orders. Beyond NLO only the two-loop virtual amplitudes in this limit are known [5]. For Standard Model Higgs pairs the corrections in the same limit are known to N^{3}LO [6–8]. Pairs of axion-like particles coupled to gluons are produced by the same mechanism, and non-resonant $gg \to aa$ production has recently been shown to constrain their interactions at the LHC [9].

When the top quark is integrated out, the pseudoscalar couples to gluons through $G\tilde{G}$ and to light quarks through the divergence of the axial current. The amplitude for $gg \to AA$ then separates into two classes [5]. In Type I both pseudoscalars originate from a single effective vertex. In Type II each pseudoscalar is emitted from its own vertex and the two vertices are connected by gluon exchange in the t or u channel (Fig. 1a). In the full theory the Type II diagrams correspond to two top quark triangles joined by a gluon, the one-particle reducible contributions of Refs. [2, 3]; for Higgs pairs the corresponding class was computed at two loops in Ref. [7]. The two-loop Type II finite remainders for $gg \to AA$ were obtained in closed form in Ref. [5]. They contain polylogarithms of weight up to four, among them $\mathrm{Li}_{2,2}$, and inverse powers of the Gram determinant $tu - m_A^4$, which vanishes at the pair threshold. The expressions occupy about 204 MB, and large cancellations between their terms make them hard to evaluate close to threshold.

There are two reasons to study this region. The first is phenomenological. In the NLO predictions of Ref. [2] for $m_A = 134.7$ GeV the invariant mass distribution of the pair is larger at $m_{AA} = 300$ GeV, about 30 GeV above threshold, than at 400 GeV, and threshold resummation of this distribution requires at each m_{AA} a hard function obtained from the infrared finite virtual amplitude [10–14]. The second concerns the analytic structure of loop amplitudes. When the transverse momentum p_T of the final state particles vanishes, a one-loop amplitude with massless internal lines can have a Landau singularity [15] in which each incoming parton splits into two collinear partons that then take part in two separate $2 \to 1$ scatterings. Gaunt and Stirling showed that this double parton scattering (DPS) singularity is at most logarithmic in p_T for amplitudes of the Standard Model type, that for the processes they considered it requires the incoming partons to be in a state with $J_z = 0$, and they gave a factorised formula for its coefficient [16]. Dixon and Esterlis studied the corresponding singularity of six-gluon amplitudes in planar $\mathcal{N} = 4$ super Yang–Mills theory to high loop order [17]. As far as I am aware, it has not been examined beyond one loop in QCD at the level of amplitudes for a $2 \to 2$ process with a colour singlet final state; next-to-leading order corrections to the underlying splitting mechanism are known for double parton distributions [18]. The Type II amplitude has this topology: each incoming gluon can split into two gluons, which fuse pairwise into the two pseudoscalars at the effective vertices

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(Fig. 1b). At threshold $p_T \to 0$ at every scattering angle, so an expansion about threshold probes this region directly. In hadronic collisions the region in which both incoming partons split overlaps with double parton scattering in which both parton pairs arise from perturbative splittings [18], and the two contributions have to be combined without double counting [19]; the related contribution with a splitting in one proton has been studied for pairs of Higgs bosons [20]. Double parton scattering has since been observed in same-sign W pair production at the LHC [21, 22], and the overlap has been re-examined with two-loop splitting kernels [23].

In this paper I expand the two-loop Type II finite remainders of Ref. [5] in the velocity β_A of the pseudoscalars at fixed scattering angle, through β_A^6. The main results are the following.

1. After the endpoint constants of the expansion are expressed in a basis of eight constants, all inverse powers of β_A and the odd powers through β_A^5 cancel, and every coefficient is even in $\cos\theta$, as required by Bose symmetry. At leading power the angle enters only through the logarithm of the transverse momentum of the pseudoscalars.

2. This logarithm is the DPS singularity. At one loop its coefficient follows the factorised formula of Ref. [16], with the helicity weights of the splitting $g \to gg$ and the opposite signs of the amplitudes for $g^+g^+ \to A$ and $g^-g^- \to A$. In the forward limit, where the momentum fractions depend on the velocity, and with the overall constant fixed at threshold, the formula reproduces the velocity dependence of the exact one-loop coefficient to 3×10^{-15} for $0.001 \le \beta_A \le 0.7$. A one-loop calculation of $gg \to HH$ through two CP-even effective vertices confirms the corresponding scalar weight, with the constant fixed from AA.

3. At two loops the equal helicity amplitude grows as $\ln^3 p_T$. The opposite helicity amplitude vanishes as β_A^2 at tree level and has no DPS logarithm at one loop, but grows as $\ln^2 p_T$ at two loops. The ratio of the leading logarithms at two and one loops is the one found to all orders in planar $\mathcal{N} = 4$ super Yang–Mills theory [17].

4. In the finite virtual corrections to the cross section the one-loop logarithm cancels. At two loops $\ln^3 p_T$ cancels, and the highest power is $\ln^2 p_T$, with coefficient $-\frac{7}{4}\pi^2 N_c^2$.

I also find that the two-loop remainders released with Ref. [5] are larger by a factor of 64 than the tree and one-loop amplitudes of the same reference, and that the released one-loop remainders omit their dependence on the renormalisation scale, so that they are correct only for $\mu_R = 1$ GeV. All results are given in a consistent normalisation. The series reproduce the unexpanded remainders with relative errors of at most 6.77×10^{-5} for $\beta_A \le 0.2$; they are released together with an evaluator [24].

The paper is organised as follows. Section II collects the conventions, the normalisation of the remainders and the counting of orders. Section III describes the expansion and its accuracy. Section IV discusses the leading power terms and their relation to the DPS singularity, and Sec. V their consequences for the cross section. The conclusions are in Sec. VI.

II. AMPLITUDES AND CONVENTIONS

A. Effective theory and diagram classes

The conventions are those of Ref. [5]. After the top quark is integrated out, the pseudoscalar couples to gluons and light quarks through

$$\mathcal{L}_{\text{eff}}^{A} = \Phi_A \left[-\frac{1}{8} C_G O_G - \frac{1}{2} C_J O_J \right], \tag{1}$$

$$O_G = \varepsilon_{\mu\nu\rho\sigma} G^{a\mu\nu} G^{a\rho\sigma}, \qquad O_J = \partial_\mu \left(\bar{\psi} \gamma^\mu \gamma_5 \psi \right), \tag{2}$$

with $C_G = -2^{5/4} a_s G_F^{1/2} \cot\beta_H$, $a_s = \alpha_s/(4\pi)$ and $C_J = O(a_s)\, C_G$ [5]. Operator mixing, ultraviolet renormalisation and the treatment of γ_5 [25] follow the original calculation; the same effective theory underlies inclusive pseudoscalar production at NNLO [26].

Type I consists of the diagrams with a single effective vertex that produces both pseudoscalars, through a contact term or an intermediate scalar; these vertices are not generated by Eq. (1) alone. Type II consists of the diagrams in which each pseudoscalar is emitted from its own vertex, with the vertices connected by gluon exchange in the t or u channel (Fig. 1). The labels Type I and Type II name diagram classes, as in Ref. [5], and are unrelated to the Yukawa types of two-Higgs-doublet models; in the full theory the Type II diagrams are the one-particle reducible diagrams of Ref. [2], two top quark triangles connected by a gluon. I consider the Type II diagrams with two O_G vertices and drop the superscript II from now on. The Type II diagrams with one O_G and one O_J vertex start at one loop, where the quark loop attached to O_J couples to the gluons. The published one-loop results of Ref. [5] for these diagrams are exactly proportional to the tree amplitude of Eq. (9) below,

$$\mathcal{M}_{i,GJ}^{(1),\text{fin}} = -48\, n_f\, \mathcal{M}_i^{(0)}, \qquad i = 1, 2, \tag{3}$$

at all phase space points. This is what one expects from the axial anomaly [27, 28]: for massless quarks the divergence of the axial current equals, at one loop, a multiple of $G\tilde{G}$, so that the one-loop matrix elements of O_J are proportional to the tree-level matrix elements of O_G. The renormalised form of this relation in dimensional regularisation, with the treatment of γ_5 used here, is given in Ref. [25]. These diagrams therefore have no structure of their own near threshold.

B. Tensor decomposition and helicity amplitudes

The amplitude is decomposed as

$$\mathcal{M}_{ab}^{\mu\nu} = \delta_{ab} \left(T_1^{\mu\nu} \mathcal{M}_1 + T_2^{\mu\nu} \mathcal{M}_2 \right), \tag{4}$$

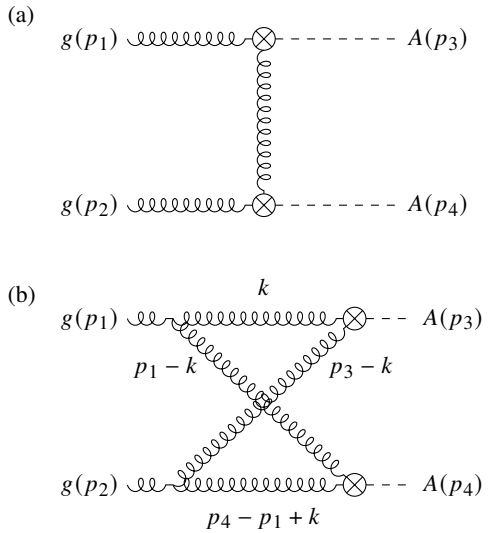


FIG. 1. (a) Type II tree diagram in the t channel; the u-channel diagram is obtained by exchanging the pseudoscalars. The crossed circles are the effective vertices of O_G. (b) One-loop Type II diagram with loop momentum k. Its double parton scattering singularity lies at $k = xp_1$ and $p_3 - k = (1-x)p_2$, where the four internal gluons are on shell and collinear with the incoming gluons: each incoming gluon splits into two, which fuse pairwise at the effective vertices. Gaunt and Stirling call this topology the crossed box [16].

with

$$T_1^{\mu\nu} = g^{\mu\nu} - \frac{p_1^\nu p_2^\mu}{p_1\cdot p_2}, \tag{5}$$

$$T_2^{\mu\nu} = g^{\mu\nu} + \frac{1}{(p_1\cdot p_2)p_T^2}\Big[m_A^2 p_2^\mu p_1^\nu - 2(p_1\cdot p_3)p_2^\mu p_3^\nu - 2(p_2\cdot p_3)p_3^\mu p_1^\nu + 2(p_1\cdot p_2)p_3^\mu p_3^\nu\Big], \tag{6}$$

$$p_T^2 = \frac{tu - m_A^4}{\hat{s}}. \tag{7}$$

For physical gluon polarisations T_1 contributes only to equal helicities ($J_z = 0$) and T_2 only to opposite helicities ($|J_z| = 2$), each with a contraction of unit modulus. Up to phases the helicity amplitudes are

$$\mathcal{A}_{\pm\pm} = C_G^2\,\mathcal{M}_1, \qquad \mathcal{A}_{\pm\mp} = C_G^2\,\mathcal{M}_2\,e^{\pm 2i\phi}, \tag{8}$$

where ϕ is the azimuth of $\vec{p}_3$, and $\sum_{\text{hel,col}}|\mathcal{A}|^2 = 2(N_c^2-1)\,C_G^4\,(|\mathcal{M}_1|^2 + |\mathcal{M}_2|^2)$. The labels 1 and 2 refer to the tensors, not to the diagram classes.

At tree level the two Type II diagrams give

$$\mathcal{M}_1^{(0)} = \frac{i\hat{s}}{2},$$
$$\mathcal{M}_2^{(0)} = \frac{im_A^2(x-z)(x^2+1)(xz-1)}{4xz(x^2-xz+1)} = 2im_A^2\beta_A^2(1-c^2) + O(\beta_A^4), \tag{9}$$

in the variables of Eq. (12). I obtained Eq. (9) from the projections of the original calculation and checked it against a direct evaluation of the two Feynman diagrams with the vertex $iC_G\varepsilon_{\mu\alpha\rho\beta}k_1^\mu k_2^\rho\delta^{ab}$ that follows from Eq. (1) (App. B). The suppression of $\mathcal{M}_2^{(0)}$ by β_A^2 is the centrifugal barrier of the $|J_z| = 2$ state.

At two loops I separate the powers of the number of light flavours,

$$\mathcal{M}_i^{(2)} = \mathcal{M}_{i,0}^{(2)} + n_f\,\mathcal{M}_{i,1}^{(2)} + n_f^2\,\mathcal{M}_{i,2}^{(2)}. \tag{10}$$

These are algebraic coefficients, not separate subprocesses. The complete coefficient below means the sum for $n_f = 5$.

C. Kinematics

The Mandelstam variables are

$$\hat{s} = (p_1+p_2)^2, \quad t = (p_1-p_3)^2, \quad u = (p_2-p_3)^2, \tag{11}$$

with $\hat{s} + t + u = 2m_A^2$. The expressions of Ref. [5] use

$$\hat{s} = m_A^2\frac{(1+x)^2}{x}, \qquad t = -m_A^2 y, \qquad u = -m_A^2 z, \tag{12}$$

with $x + x^{-1} = y + z$. With $c = \cos\theta$ in the partonic centre of mass frame,

$$\hat{s} = \frac{4m_A^2}{1-\beta_A^2}, \qquad t, u = m_A^2 - \frac{\hat{s}}{2}(1 \mp \beta_A c), \tag{13}$$

and the threshold substitution is

$$x = \frac{1-\beta_A}{1+\beta_A}, \qquad y, z = \frac{1+\beta_A^2 \mp 2\beta_A c}{1-\beta_A^2}. \tag{14}$$

All three variables tend to one as $\beta_A \to 0^+$ at fixed c, and the expansion has to follow this correlated limit. The Gram determinant and the transverse momentum are

$$tu - m_A^4 = \frac{4m_A^4\beta_A^2(1-c^2)}{(1-\beta_A^2)^2}, \qquad p_T^2 = \frac{m_A^2\beta_A^2(1-c^2)}{1-\beta_A^2}. \tag{15}$$

Both vanish at threshold at every angle, and also at fixed β_A when $c \to \pm 1$.

D. Finite remainders and normalisation

The coefficients $\mathcal{M}_i$ of Eq. (8), from which C_G^2 has been factored out, are expanded as

$$\mathcal{M}_i = \mathcal{M}_i^{(0)} + a_s\mathcal{M}_i^{(1)} + a_s^2\mathcal{M}_i^{(2)} + \cdots, \tag{16}$$

with a_s renormalised at μ_R. The finite remainders are defined by the infrared subtraction of Refs. [5, 29],

$$\mathcal{M}^{(1)} = 2I^{(1)}\mathcal{M}^{(0)} + \mathcal{M}^{(1),\text{fin}},$$
$$\mathcal{M}^{(2)} = 4I^{(2)}\mathcal{M}^{(0)} + 2I^{(1)}\mathcal{M}^{(1)} + \mathcal{M}^{(2),\text{fin}}. \tag{17}$$

From here on $\mathcal{M}^{(n)}$ denotes the finite remainder $\mathcal{M}^{(n),\mathrm{fin}}$. The remainders released with Ref. [5] are not all in the normalisation of Eq. (16). First, the two-loop remainders are larger by a factor of 64: in the two-loop renormalisation step of the original calculation the O_G–O_G amplitude, including the tree and one-loop terms that enter the subtraction, was multiplied by $64 = 8^2$. Second, the one-loop remainders do not depend on μ_R; they contain $\ln m_A^2$ where $\ln(m_A^2/\mu_R^2)$ is required. The remainders in the normalisation of Eq. (16) are therefore

$$\begin{aligned}\mathcal{M}_i^{(1)} &= \mathcal{M}_{i,\mathrm{publ}}^{(1)} + 2\beta_0 \ln\frac{\mu_R^2}{\mathrm{GeV}^2}\,\mathcal{M}_i^{(0)},\\ \mathcal{M}_i^{(2)} &= \frac{1}{64}\,\mathcal{M}_{i,\mathrm{publ}}^{(2)},\end{aligned} \tag{18}$$

with $\beta_0 = (11N_c - 2n_f)/3$ and masses in GeV in the published expressions. The one-loop term restores the scale dependence required by renormalisation group invariance; with it, the released one- and two-loop remainders are consistent with each other at every tested point. Appendix B gives the evidence. The logarithmic structure studied below, the expansion and its accuracy do not depend on these corrections; the one-loop constant and the finite virtual corrections of Sec. V do.

E. Counting of orders and range of validity

Since $C_G \propto a_s$, the Type II amplitude starts at $O(a_s^2)$, one order higher than an amplitude with a single effective vertex. In a two-Higgs-doublet model the leading order amplitude for $gg \to AA$ is of the Type I kind, and the Type II tree amplitude first contributes through its interference with it at NLO. In the NLO calculations of Refs. [2, 3] this is the one-particle reducible contribution to the virtual corrections. The two-loop Type II remainders then enter at N^{3}LO, like the corresponding class of diagrams for Higgs pairs [7]. For a pseudoscalar that couples to gluons only through $G\tilde{G}$, such as an axion-like particle without further interactions, the Type II diagrams are the complete tree amplitude and the present results are the O_G–O_G part of the two-loop virtual corrections at NNLO; in that case no heavy particle is integrated out in the amplitudes, and the results do not rely on the heavy top quark limit. Since O_G mixes with O_J under renormalisation, an O_J contribution enters at the same order; by Eq. (3) it is proportional to the tree amplitude. If dimension-six contact interactions are present, as in the analysis of Ref. [9], they play the role of Type I.

The effective theory requires the pseudoscalar and the pair invariant mass to be small compared with the top quark threshold, which near the pair threshold means $m_A \lesssim m_t$. For the Type II tree amplitude the full top quark mass dependence is known: each top quark triangle couples one pseudoscalar to an on-shell and a virtual gluon, and the amplitude of each channel is multiplied by $[I_2(\tau_A,\lambda)/I_2^\infty]^2$, with $\tau_A = 4m_t^2/m_A^2$, $\lambda = 4m_t^2/t$ or $4m_t^2/u$, and the loop function I_2 of $A \to Z\gamma$ given in Ref. [2]. At threshold, $t = u = -m_A^2$, this factor is 1.006, 1.013 and 1.022 for $m_A = 125$, 150 and 170 GeV, with $m_t = 172.5$ GeV. Near threshold the heavy top quark limit of the Type II tree amplitude is therefore accurate to about 2.2% or better for $m_A \lesssim m_t$. For the loop corrections a partial estimate follows from the factorisation discussed in Sec. IV C: the effective vertices enter the DPS logarithm only through the amplitudes for $gg \to A$ with both gluons on shell, the limit $\lambda \to \infty$ of I_2, where $I_2/I_2^\infty = \tau_A f(\tau_A)$ with $f(\tau) = \arcsin^2(1/\sqrt{\tau})$. If this factorisation carries over to the full theory, in which the top quark triangles are hard subgraphs, the coefficient of the logarithm is multiplied by $[\tau_A f(\tau_A)]^2 = 1.096$, 1.145 and 1.196 for the same masses, while its dependence on the velocity and the helicity selection rule are unchanged. I have no estimate for the remaining terms. For $m_A \gtrsim m_t$ these factors grow quickly, and the heavy top quark limit should not be used there.

III. THRESHOLD EXPANSION

A. Construction

After the substitution of Eq. (14), each flavour coefficient is expanded at fixed c,

$$\mathcal{M}_{i,r}^{(2)}(\beta_A, c) = \sum_{n=-4}^{p}\sum_k C_{nk}^{(i,r)}\,\beta_A^n \ln^k\beta_A + O_{\log}(\beta_A^{p+1}), \tag{19}$$

with $r = 0, 1, 2$ and $p = 4$ or 6. The coefficients $C_{nk}^{(i,r)}$ depend on c, m_A^2, μ_R^2 and N_c, and $O_{\log}(\beta_A^n)$ stands for β_A^n times a polynomial in $\ln\beta_A$. Inverse powers appear in intermediate steps because the tensor basis and the rational prefactors contain inverse Gram determinants, Eq. (15). They must be kept until all contributions at a given power have been combined.

The remainders contain logarithms, classical polylogarithms and the function $\mathrm{Li}_{2,2}$ of weight four [30, 31], whose arguments approach correlated endpoints as $\beta_A \to 0$. I expand the rational prefactors exactly and obtain the expansions of the special functions from differential equations in β_A, with boundary values at $\beta_A = 0$. These equations have the iterated-integral form familiar from harmonic polylogarithms [32] and from canonical differential equations for Feynman integrals [33]; the map $x = (1-\beta_A)/(1+\beta_A)$ is one of the transformations under which harmonic polylogarithms are closed [32]. The method thus expands the known remainders, not the loop integrals, which could instead be expanded near threshold by the method of regions [34]. For the dilogarithm, for example, $\mathrm{d}\,\mathrm{Li}_2(x)/\mathrm{d}\beta_A = 2(1-\beta_A^2)^{-1}\ln[2\beta_A/(1+\beta_A)]$ with $\mathrm{Li}_2(1) = \pi^2/6$ gives

$$\begin{aligned}\mathrm{Li}_2(x) = \frac{\pi^2}{6} &+ 2\beta_A\big[\ln(2\beta_A) - 1\big] - \beta_A^2\\ &+ \beta_A^3\left[\frac{2}{3}\ln(2\beta_A) + \frac{1}{9}\right] - \frac{2}{3}\beta_A^4 + O\Big(\beta_A^5\ln\beta_A\Big).\end{aligned} \tag{20}$$

Since the prefactors start at β_A^{-4}, the special functions have to be expanded four orders beyond the final accuracy. The fourth and sixth order expansions were constructed independently, and all coefficients common to both agree.

B. Endpoint constants and exact cancellations

The boundary values at $\beta_A = 0$ introduce thirteen constants $\mathrm{MB}[p, q, a, w]$, defined in Eq. (A1) of App. A. Their arguments take the values ± 1 and $\frac{1}{2}$, so they are alternating Euler sums of weight three and four [35]. An integer relation search with the PSLQ algorithm [36] at 110 digits expresses each of them in the basis

$$\{\zeta_3,\ \pi^2 \ln 2,\ \ln^3 2,\ \pi^4,\ \pi^2 \ln^2 2,\ \ln^4 2,\ \mathrm{Li}_4(\tfrac{1}{2}),\ \zeta_3 \ln 2\}, \quad (21)$$

with a largest residual of 4×10^{-107} (Table VII).

With these reductions, and with the dilogarithms of $(1 - c)/2$, $(c - 1)/(c + 1)$ and $(c + 1)/(c - 1)$ rewritten through $\mathrm{Li}_2\big((1 + c)/2\big)$ by the reflection and Landen identities, every coefficient $C_{nk}^{(i,r)}$ becomes a polynomial in

$$\Big\{\ln \beta_A,\ \ln(1 \pm c),\ \mathrm{Li}_2\big(\tfrac{1+c}{2}\big),\ \ln \tfrac{m_A^2}{\mu_R^2}\Big\} \cup \{\ln 2,\ \pi,\ \zeta_3,\ \mathrm{Li}_4(\tfrac{1}{2})\} \quad (22)$$

with coefficients rational in c and N_c. Treating these quantities as algebraically independent, I find that

1. the coefficients of $\beta_A^{-4}, \ldots, \beta_A^{-1}$ vanish identically in every flavour coefficient;
2. the coefficients of β_A, β_A^3 and β_A^5 vanish identically, so that through β_A^6 the expansion is a series in β_A^2 with logarithms;
3. the coefficients of β_A^0, β_A^2, β_A^4 and β_A^6 of all six flavour coefficients, 24 in total, are even in c, so that Bose symmetry holds order by order.

The cancellations and the reductions of the constants were confirmed independently in Mathematica.

The cancellations have a simple origin. Exchanging the pseudoscalars exchanges t and u, and the invariants that are symmetric under this exchange, $\hat{s} = 4m_A^2/(1 - \beta_A^2)$, $(t - u)^2 = \hat{s}^2 \beta_A^2 c^2$ and $tu - m_A^4 = \hat{s}\, p_T^2$ with $p_T^2 = \hat{s}\beta_A^2(1 - c^2)/4$, depend on β_A only through β_A^2. No massive particle propagates in the loops, so the remainders are regular at $\hat{s} = 4m_A^2$, and a colourless final state has no Coulomb singularity. Near threshold the only singularity is then the one at $p_T \to 0$ discussed in Sec. IV C, which is logarithmic at one loop [16]; the inverse Gram determinants of the tensor basis vanish at the same point but belong to the basis, not to the amplitude. A Bose symmetric function whose only singularity is logarithmic in p_T^2 is a series in β_A^2 with logarithms and has no inverse powers. At two loops this accounts for the cancellations found above but does not prove them, since there the logarithmic nature of the singularity is known only from the result itself. Table I lists the leading terms. Since the first omitted term is $O_{\log}(\beta_A^{p+2})$, the truncation errors should scale as β_A^6 and β_A^8 for $p = 4$ and 6.

The coefficient of n_f^2 is the simplest:

$$\mathcal{M}_{1,2}^{(2)}(\beta_A, c) = K + O(\beta_A^2), \quad (23)$$

$$\mathcal{M}_{2,2}^{(2)}(\beta_A, c) = \beta_A^2(1 - c^2)K + O(\beta_A^4), \quad (24)$$

where K does not depend on the angle. It is the only flavour coefficient that follows the tree-level pattern $\mathcal{M}_2 \propto \beta_A^2(1 - c^2)$ of Eq. (9), with the same ratio to the tree amplitude in both tensor structures. The complete expressions through β_A^6 are released as Wolfram Language files (App. D).

TABLE I. Leading terms of the threshold expansion of the two-loop Type II remainders. The second column lists the inverse powers with nonzero coefficients in the construction files; they cancel identically after the reductions of Sec. III B, as do all odd powers. The last column gives the powers of $\mathcal{L}$ at leading power.

quantity	inverse powers in construction	leading power	powers of $\mathcal{L}$
$\mathcal{M}_{1,0}^{(2)}$	$\beta_A^{-2}, \beta_A^{-1}$	β_A^0	0, 1, 2, 3
$\mathcal{M}_{1,1}^{(2)}$	–	β_A^0	0, 1, 2
$\mathcal{M}_{1,2}^{(2)}$	–	β_A^0	0
$\mathcal{M}_{2,0}^{(2)}$	$\beta_A^{-4}, \beta_A^{-3}, \beta_A^{-2}, \beta_A^{-1}$	β_A^0	0, 1, 2
$\mathcal{M}_{2,1}^{(2)}$	β_A^{-1}	β_A^0	0, 1
$\mathcal{M}_{2,2}^{(2)}$	–	β_A^2	0

C. Accuracy

I compared the series with the unexpanded remainders at 9 values of β_A between 10^{-3} and 3/10, 5 angles and 3 choices of mass, scale and colour (Table II). At each of the 135 points the test covers the six flavour coefficients and the two complete coefficients, 1080 comparisons in total. Each unexpanded value is evaluated at increasing working precision until its real and imaginary parts are stable. The parameter set with $N_c \neq 3$ tests the colour dependence; all physical results use $N_c = 3$. With the relative truncation error

$$E_i^{(p)} = \frac{\big|\mathcal{M}_i^{[p]} - \mathcal{M}_i\big|}{|\mathcal{M}_i|}, \qquad p = 4, 6, \quad (25)$$

where $\mathcal{M}_i$ is the unexpanded remainder and $\mathcal{M}_i^{[p]}$ its series through β_A^p. Fig. 2 shows the maximum over angles and parameter sets at each β_A. Fits give $E_i^{(4)} \propto \beta_A^{6.005}$ and $\beta_A^{5.955}$, and $E_i^{(6)} \propto \beta_A^{8.005}$ and $\beta_A^{8.034}$, as expected for a series in β_A^2. The lower panels divide out these powers. The ratios vary only slowly with β_A, because the first omitted coefficient contains powers of $\ln \beta_A$, and change most near $\beta_A = 0.3$, where the next order is no longer negligible; an error at a lower order would make them grow as an inverse power of β_A. For $\beta_A \leq 0.2$ the sixth order series reproduces the complete coefficients with relative errors of at most 3.23×10^{-6} and 6.77×10^{-5}; at $\beta_A = 0.3$ the errors are 1.10×10^{-4} and 1.94×10^{-3} (Table III). The values at c and $-c$ are identical in all stored digits, in 432 comparisons of the unexpanded coefficients and 864 of the series.

The reference values were computed in Mathematica from the published expressions. I also evaluated these expressions with an independent code written in Python with mpmath [37], which

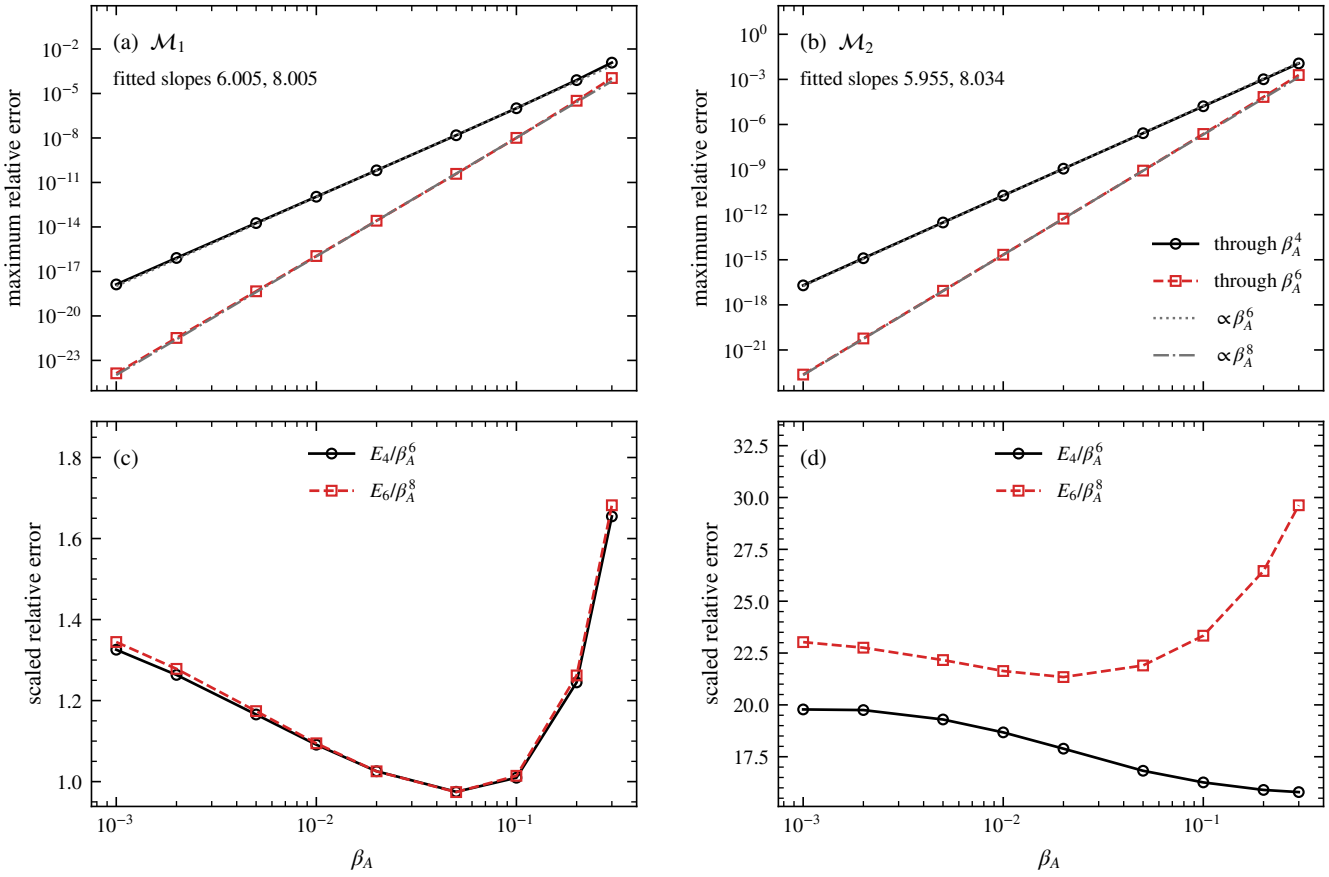


FIG. 2. Maximum relative errors of the complete two-loop coefficients after truncation at β_A^4 and β_A^6, with β_A^6 and β_A^8 for comparison (upper panels), and after division by these powers (lower panels). The maximum is taken over the five angles and three parameter sets of Table II.

TABLE II. Validation grid. At each point the six flavour coefficients and the two complete $n_f = 5$ coefficients are compared.

quantity	values
β_A	1/1000, 1/500, 1/200, 1/100, 1/50, 1/20, 1/10, 1/5, 3/10
c	$-2/3, -1/3, 0, 1/3, 2/3$
$(m_A/\text{GeV}, \mu_R/\text{GeV}, N_c)$	$(125, 125, 3)$, $(125, 250, 3)$, $(150, 75, 4)$

TABLE III. Maximum relative errors of the complete Type II coefficients on the validation grid. At each value of β_A, the maximum is taken over the 5 angles and 3 parameter sets of Table II.

	$\mathcal{M}_1$ through		$\mathcal{M}_2$ through	
β_A	β_A^4	β_A^6	β_A^4	β_A^6
0.01	1.09×10^{-12}	1.09×10^{-16}	1.87×10^{-11}	2.16×10^{-15}
0.05	1.52×10^{-8}	3.81×10^{-11}	2.63×10^{-7}	8.55×10^{-10}
0.10	1.01×10^{-6}	1.01×10^{-8}	1.63×10^{-5}	2.33×10^{-7}
0.20	7.97×10^{-5}	3.23×10^{-6}	1.02×10^{-3}	6.77×10^{-5}
0.30	1.21×10^{-3}	1.10×10^{-4}	1.15×10^{-2}	1.94×10^{-3}

has its own reader of the published files and computes $\text{Li}_{2,2}$ from its defining double sum, with a one-dimensional integral representation where the sum converges slowly. At four points covering the three parameter sets, the two complete coefficients agree with the reference values to a relative difference of at most 1.3×10^{-40}, the precision to which the latter are stored; the six flavour coefficients at $\beta_A = 1/10$, $c = 1/3$ agree to 1.6×10^{-40}. The published expressions contain removable singularities at $c = 0$, where $t = u$, which have to be taken as limits.

The expansion is carried out at fixed angle, but p_T also vanishes when $c \to \pm 1$ at fixed β_A. I therefore extended the comparison to $c = 0.9, 0.99, 0.999$ and -0.99 at $\beta_A = 0.1, 0.2$ and 0.3 (Table IV). All values are below the maximum on the main grid at the same β_A, except for $\mathcal{M}_1$ at $\beta_A = 0.1$, where the largest value, 1.23×10^{-8} at $c = 0.999$, exceeds it by 21%. The series thus remains accurate close to the beam, consistent with the absence of poles at $c = \pm 1$ in its coefficients (Sec. IV A).

TABLE IV. Relative truncation errors $E_i^{(6)}$ of the complete $n_f = 5$ coefficients towards the beam, at $m_A = \mu_R = 125\,\text{GeV}$ and $N_c = 3$. The last rows give the largest value on the grid of Table II. The values at $c = -0.99$ equal those at $-c$.

	c	$\beta_A = 0.1$	$\beta_A = 0.2$	$\beta_A = 0.3$
$\mathcal{M}_1$	0.9	6.36×10^{-9}	1.11×10^{-6}	1.78×10^{-5}
$\mathcal{M}_1$	0.99	1.06×10^{-8}	2.43×10^{-6}	5.76×10^{-5}
$\mathcal{M}_1$	0.999	1.23×10^{-8}	2.96×10^{-6}	7.18×10^{-5}
$\mathcal{M}_2$	0.9	1.39×10^{-7}	3.08×10^{-5}	6.68×10^{-4}
$\mathcal{M}_2$	0.99	6.68×10^{-8}	1.47×10^{-5}	2.88×10^{-4}
$\mathcal{M}_2$	0.999	6.21×10^{-8}	1.65×10^{-5}	3.74×10^{-4}
$\mathcal{M}_1$	grid max.	1.01×10^{-8}	3.23×10^{-6}	1.10×10^{-4}
$\mathcal{M}_2$	grid max.	2.33×10^{-7}	6.77×10^{-5}	1.94×10^{-3}

IV. LEADING POWER AND THE DOUBLE PARTON SCATTERING SINGULARITY

A. Leading power

The leading power coefficients depend on $\ln \beta_A$ and on $\ln(1 \pm c)$. When they are written in terms of

$$\begin{aligned}\mathcal{L} &= \ln \beta_A + \tfrac{1}{2} \ln(1 - c^2) + \ln 2 + \frac{i\pi}{2} \\ &= \tfrac{1}{2} \ln\bigl[-4\beta_A^2 (1 - c^2)\bigr] = \tfrac{1}{2} \ln \frac{m_A^4 - tu}{m_A^4} + O(\beta_A^2) \\ &= \ln \frac{2p_T}{m_A} + \frac{i\pi}{2} + O(\beta_A^2),\end{aligned} \tag{26}$$

the dependence on the angle disappears: no $\ln(1 \pm c)$ or $\mathrm{Li}_2\bigl((1 + c)/2\bigr)$ remains outside $\mathcal{L}$, and no rational function of c remains. For general N_c and n_f,

$$\begin{aligned}\mathcal{M}_1^{(2)}\big|_{\beta_A^0} &= m_A^2 \sum_{k=0}^{3} \widehat{A}_{1k}\, \mathcal{L}^k, \\ \mathcal{M}_2^{(2)}\big|_{\beta_A^0} &= m_A^2 \sum_{k=0}^{2} \widehat{A}_{2k}\, \mathcal{L}^k,\end{aligned} \tag{27}$$

with

$$\begin{aligned}\widehat{A}_{13} &= -\tfrac{28}{3} \pi N_c^2, \qquad & \widehat{A}_{12} &= -\tfrac{14}{3} \pi N_c (11 N_c - 2 n_f), \\ \widehat{A}_{22} &= -2\pi N_c^2, & \widehat{A}_{21} &= -\tfrac{4}{3} \pi N_c (4 N_c - n_f),\end{aligned} \tag{28}$$

which do not depend on μ_R, and

$$\widehat{A}_{11} = \frac{\pi N_c}{9} \bigl(N_c\, a_\mathrm{A} - n_f\, a_\mathrm{F}\bigr), \tag{29}$$

where

$$\begin{aligned}a_\mathrm{A} &= 1454 + 462 \ln 2 + 252 \ln^2 2 - 42\pi^2 + 693\, i\pi - 693 L_\mu, \\ a_\mathrm{F} &= 152 + 84 \ln 2 + 126\, i\pi - 126 L_\mu,\end{aligned}$$

and $L_\mu = \ln(m_A^2/\mu_R^2)$. Table V gives all coefficients for $N_c = 3$, $n_f = 5$ and $\mu_R = m_A$. At β_A^2, β_A^4 and β_A^6 the same substitution again removes every $\ln(1 \pm c)$ and Li_2; the remaining dependence on the angle is rational, even in c and free of poles at $c = \pm 1$.

Up to the constant $i\pi/2$ and power corrections, $\mathcal{L}$ is the logarithm of the transverse momentum of the pseudoscalars or, equivalently, of the Gram determinant $tu - m_A^4$. The exact one-loop remainders already contain it, since they depend on $\ln(y-x) + \ln(z-x)$ and $(y-x)(z-x) = yz - 1 = (tu - m_A^4)/m_A^4$ by Eq. (12). The two-loop coefficients have no finite limit as $\beta_A \to 0$; they grow as $\mathcal{L}^3$ and $\mathcal{L}^2$.

B. One loop

Fitting the published one-loop remainders at $\beta_A = 10^{-40}$ and 10^{-50} with 400-digit arithmetic, for $N_c = 3$, 4 and 5, $n_f = 0$, 2 and 5, two values of μ_R and several angles, I find to 30 digits

$$\begin{aligned}\mathcal{M}_1^{(1)} &= m_A^2 \bigl[7\pi N_c\, \mathcal{L} + \widehat{A}_{10}^{(1)}\bigr] + O(\beta_A), \\ \mathcal{M}_2^{(1)} &= \tfrac{1}{2} \pi N_c\, m_A^2 + O(\beta_A),\end{aligned} \tag{30}$$

with $\widehat{A}_{10}^{(1)} = -93.900 + 11.484\, i$ for $N_c = 3$, $n_f = 5$ and $\mu_R = m_A$, in the normalisation of Eq. (18). The constant does not depend on the angle, and by Eq. (18) its scale dependence is $\partial \widehat{A}_{10}^{(1)} / \partial \ln \mu_R^2 = 4i\beta_0$. The coefficient of $\mathcal{L}$ and the limit of $\mathcal{M}_2^{(1)}$ depend neither on n_f nor on μ_R. Together with Eqs. (9) and (27), the behaviour at threshold is summarised in Table VI.

The $|J_z| = 2$ amplitude is suppressed near threshold only at tree level. According to Eq. (8) the helicity amplitude $\mathcal{A}_{\pm\mp}$ then keeps its dependence $e^{\pm 2i\phi}$ on the azimuth of $\vec{p}_3$ as $|\vec{p}_3| \to 0$. An amplitude that is analytic in $\vec{p}_3$ at $\vec{p}_3 = 0$ cannot behave in this way; the loop amplitudes can, because they depend on $\ln p_T$.

C. The double parton scattering singularity

The origin of $\mathcal{L}$ can be identified. At one loop the Type II amplitude contains diagrams in which each incoming gluon splits into two gluons through a three-gluon vertex, and the four gluons fuse pairwise at the two effective vertices into the pseudoscalars. When $p_T \to 0$ the four internal propagators can go on shell simultaneously, with one pair collinear to each incoming gluon. This configuration gives the pinch singularity that Gaunt and Stirling identified as the DPS singularity of one-loop amplitudes [16]. For theories of the Standard Model type they showed that it is at most logarithmic in p_T, that in the processes they considered, with fermions in the loop, it appears only if the incoming partons are in a state with $J_z = 0$, and that its coefficient is a transverse momentum integral of light-cone wavefunctions of the incoming partons and tree-level matrix elements of the two $2 \to 1$ subprocesses. In their analysis the strength of the singularity is set by the vertices at which the incoming partons split, whereas the $2 \to 1$ amplitudes enter only through their values for collinear on-shell partons at zero transverse momentum. Here the splittings occur at ordinary three-gluon vertices and the effective vertices of O_G

TABLE V. Coefficients of the leading power term, $\mathcal{M}_i|_{\beta_A^0} = m_A^2 \sum_k \widehat{A}_{ik}\, \mathcal{L}^k$, with $\mathcal{L}$ of Eq. (26). They do not depend on the scattering angle. Exact forms for $\mu_R = m_A$ and general N_c, n_f; numerical values for $N_c = 3$, $n_f = 5$, $\mu_R = m_A$.

	k	exact	Re $\widehat{A}_{ik}$	Im $\widehat{A}_{ik}$
$\mathcal{M}_1$	3	$-\frac{28}{3}\pi N_c^2$	-2.63894×10^2	0
$\mathcal{M}_1$	2	$-\frac{14}{3}\pi N_c(11N_c - 2n_f)$	-1.01159×10^3	0
$\mathcal{M}_1$	1	Eq. (29)	3.55129×10^3	4.76702×10^3
$\mathcal{M}_1$	0	–	-3.98215×10^2	-4.00569×10^3
$\mathcal{M}_2$	2	$-2\pi N_c^2$	-5.65487×10^1	0
$\mathcal{M}_2$	1	$-\frac{4}{3}\pi N_c(4N_c - n_f)$	-8.79646×10^1	0
$\mathcal{M}_2$	0	–	3.12007×10^2	3.40501×10^2

TABLE VI. Behaviour of the Type II coefficients at threshold, $\beta_A \to 0$ at fixed angle: tree-level values, and the highest power of $\mathcal{L}$ at one and two loops.

	tree level	one loop	two loops
$\mathcal{M}_1$ ($J_z = 0$)	$2im_A^2$	$\mathcal{L}$	$\mathcal{L}^3$
$\mathcal{M}_2$ ($\lvert J_z\rvert = 2$)	$2im_A^2\beta_A^2(1-c^2)$	$\mathcal{L}^0$	$\mathcal{L}^2$

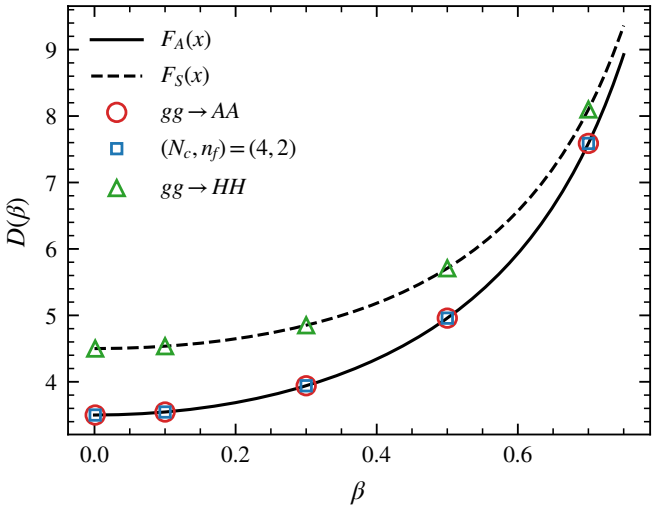


FIG. 3. Forward coefficient $D(\beta)$ of $\ln p_T^2$ in the one-loop remainder, Eq. (33), against the velocity β of the produced bosons. Circles: $gg \to AA$ with $(N_c, n_f) = (3, 5)$; squares: $(N_c, n_f) = (4, 2)$; the two coincide. Triangles: $gg \to HH$ through two CP-even vertices at $(N_c, n_f) = (3, 5)$, multiplied by -1, the sign of its tree amplitude at threshold relative to AA. Lines: the helicity weights F_A (solid) and F_S (dashed) of Eq. (32) at $x = (1 + \beta)/2$.

enter as the amplitudes for $gg \to A$, so the analysis applies. For $g \to gg$ every collinear splitting changes J_z, since the total helicity of a collinear pair of gluons is ± 2 or 0, never that of the parent gluon, so the splitting amplitudes vanish linearly in the transverse momentum and the singularity is logarithmic, as for loops of massless quarks [16]. With gluons in the loop, helicity conservation no longer restricts the four partons of the singular configuration to $J_z = 0$, and the requirement is conservation of J_z between the incoming partons and these four partons [17]. For the present process this again selects $J_z = 0$. The amplitude for $gg \to A$ vanishes for opposite gluon helicities, so each pair of gluons that fuses at an effective vertex has $J_z = 0$, and so does the configuration as a whole; the singularity can therefore appear only for incoming gluons with $J_z = 0$.

In the present case the subprocesses are $gg \to A$ through O_G. Their amplitudes vanish for opposite gluon helicities, and, since a parity transformation reverses all helicities [38], a CP-odd coupling gives amplitudes of opposite sign for $g^+g^+ \to A$ and $g^-g^- \to A$; for a scalar coupling to G^2 they would be equal. For the splitting of a gluon of positive helicity into gluons a and b with momentum fractions z and $1 - z$, the splitting amplitudes are proportional to [38, 39]

$$\frac{1}{\sqrt{z(1-z)}}, \quad \frac{z^2}{\sqrt{z(1-z)}}, \quad \frac{(1-z)^2}{\sqrt{z(1-z)}} \tag{31}$$

for the helicities $(h_a, h_b) = (+,+)$, $(+,-)$ and $(-,+)$; the sum of their squares gives the numerator $1 + z^4 + (1-z)^4$ of the $g \to gg$ splitting function. Let the first pseudoscalar receive the fraction x of the momentum of the first gluon and the fraction $1 - x$ of the momentum of the second, as for the kinematics of Ref. [16] with equal masses; in Fig. 1(b) this is the point $k = xp_1$, $p_3 - k = (1-x)p_2$. Summing over the three helicity configurations allowed by the subprocesses, with the signs of their amplitudes, gives the weight

$$F_A(x) = \frac{1}{x(1-x)} - 2x(1-x), \tag{32}$$

where the first term comes from $(+,+)$ splittings of both gluons and the second from the two mixed configurations, in which one effective vertex receives two gluons of positive and the other two of negative helicity, so that this term carries the relative sign of the two amplitudes. A scalar would have $F_S(x) = 1/[x(1-x)] + 2x(1-x)$ instead. Boxes of massless quarks have a different weight: for $gg \to HH$ through a quark loop the prefactor is proportional to $x(1-x)$, because the internal lines of equal helicity carry complementary momentum fractions [16].

At threshold both pseudoscalars are at rest and $x = \frac{1}{2}$, so that $F_A = \frac{7}{2}$ and $F_S = \frac{9}{2}$. The coefficient $7\pi N_c$ of $\mathcal{L}$ in Eq. (30) corresponds to $\partial \mathcal{M}_1^{(1)}/\partial \ln p_T^2 = \frac{7}{2}\pi N_c m_A^2$, that is $\pi N_c m_A^2 F_A(\frac{1}{2})$. A test that does not rely on the overall factor is provided by the forward limit $c \to 1$ at fixed velocity, where

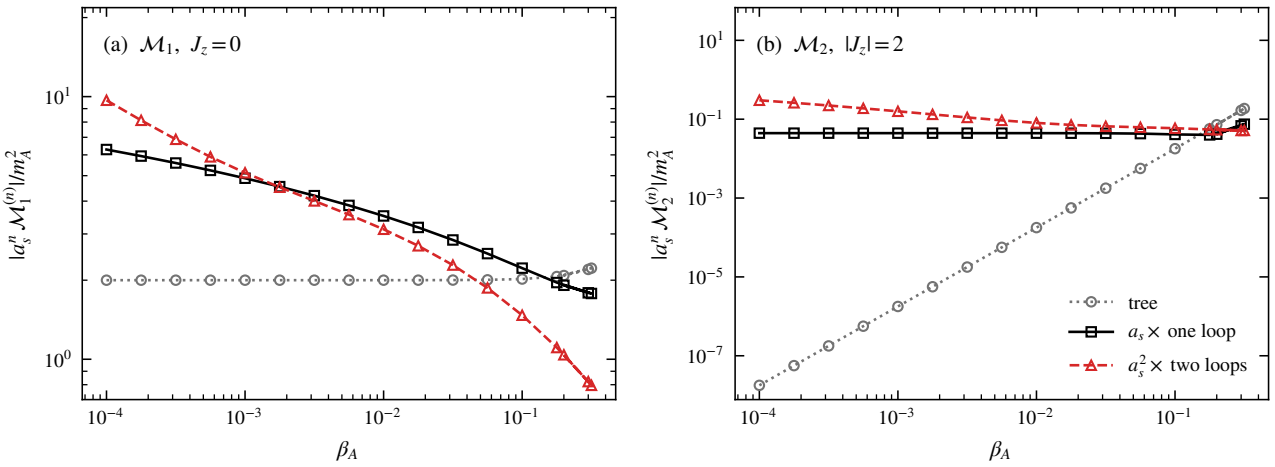


FIG. 4. Tree-level, one-loop and two-loop terms of the Type II amplitude, $|a_s^n \mathcal{M}_i^{(n)}|/m_A^2$ with $a_s = 0.0094$ ($\alpha_s = 0.118$), at $c = 1/3$, $m_A = \mu_R = 125\,\text{GeV}$, $N_c = 3$ and $n_f = 5$. Tree level: Eq. (9); one loop: published remainder; two loops: series through β_A^6. $\mathcal{M}_1$ and $\mathcal{M}_2$ are the equal and opposite helicity amplitudes, Eq. (8).

$p_T \to 0$ as well but the bosons move along the beam. Writing β and m for the velocity and mass of the produced bosons ($\beta = \beta_A$ and $m = m_A$ for AA), so that $x = (1+\beta)/2$, the DPS singularity requires for the forward coefficient $D(\beta)$

$$D(\beta) \equiv \frac{1}{\pi N_c m^2} \frac{\partial\, \text{Re}\, \mathcal{M}_1^{(1)}}{\partial \ln p_T^2}\bigg|_{c\to 1} = F_A\left(\frac{1+\beta}{2}\right). \qquad (33)$$

I evaluated D from the exact one-loop remainders at $1-c = 10^{-14}$ and 10^{-18}, for $0.001 \le \beta \le 0.7$, $N_c = 3$ and 4 and $n_f = 0$, 2 and 5. Equation (33) holds in all cases with a largest relative deviation of 3×10^{-15} (Fig. 3). The weight of a scalar is excluded at every $\beta > 0$. The overall factor $\pi N_c m^2$ is fixed by the threshold value; I have not derived it from the vertices.

The identification explains the other features of Eq. (30). The $|J_z| = 2$ amplitude $\mathcal{M}_2$ has no logarithm at one loop, as the J_z argument above requires. The coefficient of $\mathcal{L}$ does not depend on n_f, because the vertices of O_G couple to gluons only, so that only gluon splittings enter the singular configuration. It does not depend on μ_R, because the singular term is finite and receives no contribution from renormalisation or from the infrared subtraction. Its colour factor N_c is that of the two three-gluon vertices. Finally, the coefficient $7\pi N_c\, m_A^2$ is real while the tree amplitude $\mathcal{M}_1^{(0)} = 2im_A^2$ is imaginary, so the logarithm is out of phase with the tree amplitude by 90°. This holds for any real couplings: the DPS singularity of a one-loop amplitude comes from the cuts that put the internal lines on shell and is therefore absorptive [16]. This has direct consequences for the cross section (Sec. V).

A CP-even coupling, $HG^a_{\mu\nu}G^{a\mu\nu}$, gives equal amplitudes for $g^+g^+ \to H$ and $g^-g^- \to H$, which reverses the sign of the mixed term and gives the weight F_S. The topology, the splittings and the colour factors are the same as for AA, so HH tests the part of the formula that depends on the vertices. I tested this prediction with the one-loop Type II amplitudes for $gg \to HH$ at the same mass m, computed with the code of Ref. [5] after replacing the effective vertices by the CP-even ones; the code reproduces the one-loop AA result of that reference to 40 digits, and the HH double poles are those of Catani's formula [29]. The HH tree amplitude at threshold, $-2im^2$, has the sign opposite to that of AA, and $D(\beta) = -F_S\big((1+\beta)/2\big)$ to 3×10^{-15} for $0.001 \le \beta \le 0.7$ (Fig. 3). In units of the tree amplitude, the normalisation fixed from AA thus predicts the HH logarithm without a free parameter, in the ratio $F_S(\frac{1}{2})/F_A(\frac{1}{2}) = \frac{9}{7}$ at threshold. The $|J_z| = 2$ amplitude again has no logarithm; its forward coefficient is below 3×10^{-14} of that of $\mathcal{M}_1$. The diagram with the four-gluon term of the CP-even operator and the contact counterterm of two such vertices, neither of which depends on p_T, were not included.

D. Two loops

At two loops the leading logarithms of $\mathcal{M}_1$ are related to the one-loop DPS logarithm,

$$\widehat{A}_{13}\mathcal{L}^3 + \widehat{A}_{12}\mathcal{L}^2 = 7\pi N_c\, \mathcal{L}\left[-\tfrac{4N_c}{3}\,\mathcal{L}^2 - 2\beta_0\,\mathcal{L}\right], \qquad (34)$$

with $\beta_0 = (11N_c - 2n_f)/3$. In planar $\mathcal{N} = 4$ super Yang–Mills theory the singular terms of the MHV six-gluon amplitude near the configuration that mimics double parton scattering obey an evolution equation, and the leading logarithms are known to all orders [17]. There the ratio of the two- and one-loop leading logarithms is $-(a/6)\ln^2\delta$, with $a = g^2N_c/(8\pi^2)$ and δ proportional to the square of the transverse momentum. With $a = 2N_c a_s$ and $\ln\delta = 2\mathcal{L} + \text{const}$ this is $-\frac{4}{3}N_c a_s \mathcal{L}^2$, the ratio of the first terms in Eq. (34). The second term, $-2\beta_0\mathcal{L}$, has no counterpart there, since the β function of that theory vanishes, and the number of quark flavours enters it only through β_0. The same β_0 term appears in the momentum-space kernel for the splitting $g \to gg$ in unpolarised colour singlet double parton distributions at next-to-leading order [18], where the transverse momentum mismatch Δ between amplitude and conjugate plays the role of p_T. Relative to the leading order term, the β_0 part of the coefficient of $\ln^2(\mu^2/\Delta^2)$ corresponds to $-2\beta_0\mathcal{L}$. One half comes from the renormalisation of the coupling and the

other half from the δ-function parts of the DGLAP kernels, so that the running of the coupling alone would give $-\beta_0\mathcal{L}$. The remaining terms, one proportional to $C_A \ln[x(1-x)]$ from the plus distributions, which contributes where the two momentum fractions add up to one, and the real emission terms, which contribute where they add up to less than one, have no counterpart in the amplitude. The $|J_z| = 2$ amplitude acquires a logarithm at two loops, $\widehat{A}_{22} = (-4N_c) \cdot \frac{1}{2}\pi N_c$, that is the one-loop constant multiplied by $-4N_c\,\mathcal{L}^2$. The J_z argument of Sec. IV C concerns the one-loop singularity and does not forbid this. The logarithm cannot come from two-loop amplitudes of the subprocesses, since conservation of angular momentum forbids $gg \to A$ for opposite gluon helicities to all orders, so the one-loop mechanism does not explain it; I leave its origin open. A related effect occurs in planar $\mathcal{N} = 4$ super Yang–Mills theory: the NMHV six-gluon amplitude, which has no such singularity at one loop because of J_z conservation, has logarithmic singularities from two loops on, found where the limit overlaps with multi-Regge kinematics, and Dixon and Esterlis suggest that they may come from contributions in which more than one particle crosses a cut [17].

Figure 4 compares the three orders. With $a_s = 0.0094$ ($\alpha_s = 0.118$), at $\beta_A = 0.1$ and $c = 1/3$, the one- and two-loop terms of $\mathcal{M}_1$ are 1.1 and 0.7 times the tree term, and the one-loop term of $\mathcal{M}_2$ is 2.3 times the tree term. The last ratio grows as β_A^{-2} towards threshold, since the tree-level $|J_z| = 2$ amplitude vanishes there. The one-loop terms include the scale term of Eq. (18). Towards threshold the loop terms of $\mathcal{M}_1$ grow with $|\mathcal{L}|$, the two-loop term as $|\mathcal{L}|^3$ and the one-loop term as $|\mathcal{L}|$, so that below $\beta_A \approx 1.7 \times 10^{-3}$ the two-loop term exceeds the one-loop term at this angle. Because the logarithms are out of phase with the tree amplitude, this growth largely drops out of the cross section (Sec. V). In panel (b) the one-loop term of $\mathcal{M}_2$ tends to the constant of Eq. (30), while the two-loop term grows as $\mathcal{L}^2$, the logarithm that the $|J_z| = 2$ amplitude acquires at two loops.

V. CONSEQUENCES FOR THE CROSS SECTION

A. Leading order

Averaging Eq. (8) over helicities and colours, and including a factor $\frac{1}{2}$ for identical particles, gives the leading order partonic cross section of the Type II amplitude,

$$\frac{\mathrm{d}\hat{\sigma}}{\mathrm{d}c} = \frac{\beta_A\, C_G^4}{128\pi\hat{s}\,(N_c^2-1)} \left(|\mathcal{M}_1^{(0)}|^2 + |\mathcal{M}_2^{(0)}|^2\right) \tag{35}$$

for $-1 < c < 1$. For $\alpha_s = 0.118$, $\cot\beta_H = 1$ and $m_A = 125\,\mathrm{GeV}$ I find $\hat{\sigma} = 1.29 \times 10^{-8}$ pb at $\beta_A = 0.1$ (Fig. 5(a)). The cross section rises as β_A above threshold and as $\hat{s}$ at large $\hat{s}$, as expected for dimension-five operators, and it is of fourth order in a_s through C_G^4. In a two-Higgs-doublet model it is a small part of the total, which is dominated by Type I (Sec. II E).

B. Finite virtual corrections

With all remainders in one normalisation,

$$\sum |\mathcal{A}|^2 \propto \sum_i |\mathcal{M}_i^{(0)}|^2 \left[1 + a_s V_1 + a_s^2 V_2\right],$$
$$V_1 = \frac{2\,\mathrm{Re}\sum_i \mathcal{M}_i^{(0)*}\mathcal{M}_i^{(1)}}{\sum_i |\mathcal{M}_i^{(0)}|^2},$$
$$V_2 = \frac{\sum_i \left(2\,\mathrm{Re}\ \mathcal{M}_i^{(0)*}\mathcal{M}_i^{(2)} + |\mathcal{M}_i^{(1)}|^2\right)}{\sum_i |\mathcal{M}_i^{(0)}|^2}. \tag{36}$$

I average over the angle with the leading order weight and evaluate $\mathcal{M}^{(1)}$ exactly and $\mathcal{M}_i^{(2)}$ from the series through β_A^6, both in the normalisation of Eq. (18). For $\alpha_s = 0.118$ and $\mu_R = m_A = 125\,\mathrm{GeV}$, $a_s V_1$ lies between 1.06 and 1.08 and $a_s^2 V_2$ between 0.79 and 1.03 for $10^{-3} \le \beta_A \le 0.3$ (Fig. 5(b)). V_1 and V_2 are finite parts after the subtraction of Eq. (17) and depend on that scheme; they are neither observables nor K factors. By Eq. (18), V_1 depends on the scale as $\partial V_1/\partial \ln\mu_R^2 = 4\beta_0$.

Near threshold the logarithms of Sec. IV enter the cross section in a simple way. Since $\mathcal{M}_1^{(0)} \to 2im_A^2$ and $|\mathcal{M}_2^{(0)}| \propto \beta_A^2$, Eq. (30) gives

$$V_1 \;\longrightarrow\; \frac{\pi}{2} \cdot 7\pi N_c + \mathrm{Im}\,\widehat{A}_{10}^{(1)} = 115.11 \tag{37}$$

for $\mu_R = m_A$. The DPS logarithm drops out, because it is out of phase with the tree amplitude, and only the constant $i\pi/2$ contained in $\mathcal{L}$ contributes, together with the constant $\widehat{A}_{10}^{(1)}$. V_1 therefore approaches a constant, as seen in Fig. 5. At two loops, write $\mathcal{L} = \ell + i\pi/2$ with $\ell = \ln(2p_T/m_A)$. At fixed angle the leading power terms give

$$V_2 = v_0 + v_1\,\ell + v_2\,\ell^2, \tag{38}$$

and ℓ^3 cancels because $\widehat{A}_{13}$ is real; $\widehat{A}_{12}$ is real as well and does not contribute to ℓ^2. The coefficient of ℓ^2 receives $\frac{3}{2}\pi\widehat{A}_{13} = -14\pi^2 N_c^2$ from the interference of the tree and two-loop amplitudes and $\frac{1}{4}(7\pi N_c)^2 = \frac{49}{4}\pi^2 N_c^2$ from the square of the one-loop amplitude, so that

$$v_2 = -\tfrac{7}{4}\pi^2 N_c^2. \tag{39}$$

A fit of V_2 at $c = 1/3$ and $10^{-8} \le \beta_A \le 10^{-6}$ gives $v_2 = -155.446269$, against -155.446269 from Eq. (39), a relative difference of 3×10^{-11}. The DPS logarithms thus reach the finite virtual corrections only at two loops, where $\ln^2 p_T$ is the highest power; with $a_s = 0.0094$, $a_s^2 v_2 = -0.0137$. In the language of double parton scattering the two contributions to v_2 are of different kinds. The square $|\mathcal{M}_i^{(1)}|^2$ has the DPS region on both sides of the final-state cut, like the contribution with a splitting in both protons that overlaps with double parton scattering and is removed from it by a subtraction [19, 23]. The interference with the tree amplitude has the DPS region on one side only. It carries no logarithm at one loop, because the DPS singular part is absorptive, and at two loops it is the

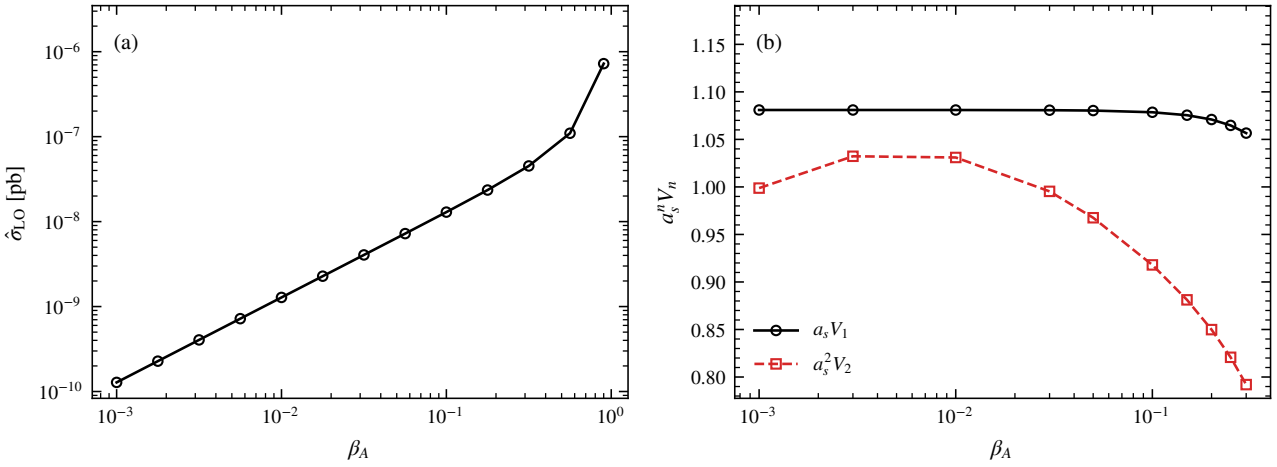


FIG. 5. (a) Leading order partonic cross section of the Type II amplitude, Eq. (35), for $m_A = 125\,\text{GeV}$, $\alpha_s = 0.118$ and $\cot\beta_H = 1$. (b) Finite virtual corrections of Eq. (36), averaged over the angle, for $\mu_R = m_A$, $N_c = 3$ and $n_f = 5$.

larger of the two and fixes the sign of v_2. At leading power v_0 and v_1 do not depend on the angle either, since the angle enters only through ℓ; fits at $c = 0$, $\frac{1}{3}$ and $\frac{2}{3}$ agree to 5×10^{-11}. Equations (28)–(30) give $v_1 = \pi\widehat{A}_{12} + \text{Im}\,\widehat{A}_{11} + \frac{7}{2}\pi N_c\,\text{Re}\,\widehat{A}_{10}^{(1)} = \frac{7}{2}\pi N_c\,\text{Re}\,\widehat{A}_{10}^{(1)} + 7\pi^2 N_c\beta_0 = -1508.5$, in agreement with the fit to 1×10^{-10}. It does not depend on μ_R, because $\text{Re}\,\widehat{A}_{10}^{(1)}$ does not; the scale dependence of V_2 sits in v_0, which changes from 8118 to 15421 when μ_R is doubled. Although ℓ^2 is the highest power, it is not the largest term at these velocities: at $c = 1/3$ the logarithmic terms $v_1\ell + v_2\ell^2$ make up 20% of V_2 at $\beta_A = 0.1$ and 30% at $\beta_A = 0.01$, of which $v_2\ell^2$ contributes -4% and -21%. Since v_1 and v_2 are both negative, V_2 has a maximum at $\ell = -v_1/(2v_2) = -4.85$, that is at $\beta_A = 0.0041$ for $c = 1/3$, and decreases above it; the angle averaged $a_s^2 V_2$ of Fig. 5(b) falls for the same reason. The terms beyond leading power change V_2 by only 0.3% at $\beta_A = 0.1$. The coefficient v_2 does not depend on the conventions of the infrared subtraction: a change of the finite parts of the operators in Eq. (17), which depend on $\hat{s}$ only, shifts the remainders by lower order amplitudes multiplied by angle independent coefficients, and so changes only v_0 and v_1.

The one-loop O_G–O_J diagrams contribute to the cross section at the same order as V_2, through the physical coefficient C_J, which depends on m_t [5]. By Eq. (3) this contribution is proportional to the tree amplitude, so it shifts V_2 by a constant; it is not included here.

C. Use in higher order calculations

Since $\sqrt{\hat{s}}/(2m_A) = (1-\beta_A^2)^{-1/2}$, the range $\beta_A \le 0.2$ covers the first 2.06% above threshold in $\sqrt{\hat{s}}$, that is $\sqrt{\hat{s}} \le 255.2\,\text{GeV}$ for $m_A = 125\,\text{GeV}$; $\beta_A = 0.3$ lies 4.83% above threshold. I recommend the series through β_A^6 for $\beta_A \le 0.2$ at the angles tested, those of Table II and the forward angles of Table IV; the logarithms diverge at $c = \pm1$, where no bound has been established. At $\beta_A = 0.3$ the series still reproduce both coefficients to 1.94×10^{-3}. The series retain the complex values of both tensor coefficients and can be combined coherently with other contributions once normalisation, Wilson coefficients and tensor basis are matched.

A threshold resummed or soft-virtual prediction of the invariant mass distribution of the pair requires, in addition to universal soft and collinear functions, the hard function of the process at each value of m_{AA}: the infrared subtracted virtual amplitude, weighted with the tree amplitude and averaged over the angle [10–12]. For Higgs pairs this is the construction of Ref. [13], in which the two-loop virtual correction enters integrated over t at fixed invariant mass. The soft and collinear functions of gluon fusion are the same for any colour singlet final state [14]. For m_{AA} within a few per cent of $2m_A$ the present series give the Type II two-loop contribution to this hard function in closed form. It enters at the order at which the remainders enter the cross section (Sec. II E): at NNLO for a pseudoscalar that couples to gluons only through $G\tilde{G}$, and at N^{3}LO in a two-Higgs-doublet model. For Higgs pairs, likewise, the NNLO virtual corrections contain the diagrams with two effective vertices only at tree level and at one loop [40], and their two-loop terms are an ingredient of N^{3}LO [7].

VI. CONCLUSIONS

I have expanded the two-loop Type II finite remainders for $gg \to AA$ about the pair threshold through β_A^6 at fixed scattering angle. After the endpoint constants are expressed in a basis of eight transcendental constants, all inverse powers of β_A and the odd powers through β_A^5 cancel identically, and every coefficient is even in $\cos\theta$. The series reproduce the unexpanded remainders to 3.23×10^{-6} and 6.77×10^{-5} for $\beta_A \le 0.2$, including the forward angles tested. They are faster to evaluate than the published expressions and 5061 times smaller than the published files. I also found that the released two-loop remainders are larger by a factor of 64 than the tree and one-loop amplitudes, and that the released one-loop remainders omit their dependence on the renormalisation scale. With both corrected, the one- and two-loop remainders satisfy

the renormalisation group relations exactly.

At leading power the angular dependence enters only through the logarithm of the transverse momentum of the pseudoscalars. I identified this logarithm with the double parton scattering singularity of Gaunt and Stirling. At one loop the velocity dependence of its coefficient follows their factorised formula, with the helicity weights of gluon splitting and the sign structure of a CP-odd coupling, to 3×10^{-15} over the range of velocities tested, once the overall constant is fixed at threshold. The same identification explains why the $|J_z| = 2$ amplitude has no logarithm at one loop, why the coefficient depends neither on n_f nor on μ_R, and why the logarithm is out of phase with the tree amplitude. At two loops the $J_z = 0$ amplitude grows as $\ln^3 p_T$ and the $|J_z| = 2$ amplitude as $\ln^2 p_T$, and the leading logarithms are simple multiples of the one-loop terms. The ratio of the leading logarithms is the one found to all orders in planar $\mathcal{N} = 4$ super Yang–Mills theory, and the next logarithm has the β_0 structure of the next-to-leading order splitting kernel for $g \to gg$. In the finite virtual corrections to the cross section the logarithm cancels at one loop; at two loops $\ln^3 p_T$ cancels and the highest power is $\ln^2 p_T$, with coefficient $-\frac{7}{4}\pi^2 N_c^2$.

Several questions remain open. The overall normalisation $\pi N_c m^2$ of the one-loop DPS logarithm, which is the same for AA and HH in units of the tree amplitude, should follow from the factorised formula of Ref. [16] with the effective vertices. That formula reproduces the numerical prefactors of the quark boxes, so this is a well-defined calculation. The two-loop logarithms, including those of the $|J_z| = 2$ amplitude, require its extension beyond one loop; the evolution equation of planar $\mathcal{N} = 4$ super Yang–Mills theory [17] and the next-to-leading order splitting kernels of Ref. [18] suggest the form such an extension might take. If the factorisation carries over to the full theory, the top quark mass changes the normalisation of the DPS logarithm by the on-shell vertex factor of Sec. II E, about 1.096 at $m_A = 125$ GeV; its effect on the remaining terms is not known. On the phenomenological side, the contribution of the physical coefficient C_J, which depends on m_t, shifts V_2 by a constant that is not included here, and the series provide the Type II two-loop contribution to the hard function for a threshold resummed description of pseudoscalar pairs near $m_{AA} \approx 2m_A$. For a pseudoscalar that couples to gluons only through $G\tilde{G}$, such as an axion-like particle, they give the O_G–O_G part of the two-loop virtual corrections at NNLO; combined with the soft and collinear functions of Refs. [12, 14], and together with that term, they would give a soft-virtual NNLO prediction for the invariant mass distribution of such particles near threshold. For Higgs pairs produced through two effective vertices, an explicit one-loop calculation confirms the scalar weight F_S, with the constant fixed from AA, and the absence of a one-loop logarithm in the $|J_z| = 2$ amplitude (Sec. IV C). The two-loop amplitudes of this class are known [7], so the two-loop logarithms can be examined as well.

DATA AVAILABILITY

The threshold series, construction files, validation records, the code of the construction and validation, the evaluator, the one-loop $gg \to HH$ coefficients of Sec. IV C and the scripts that reproduce all tables and figures are provided as ancillary files with the arXiv version of this article and as Supplemental Material [24]. The unexpanded finite remainders used as input were published with Ref. [5]. The code used for the one-loop $gg \to HH$ amplitudes of Sec. IV C is not included; it is available from the author on request.

TABLE VII. Reductions of the endpoint constants of Eq. (A1), found with PSLQ at 110 digits; the largest residual is 4×10^{-107}.

$[p, q, a, w]$	reduction
$[1, 2, -1, 1]$	$-\frac{\pi^2 \ln 2}{6} + \frac{5\zeta_3}{8}$
$[1, 2, 1/2, 1/2]$	$-\frac{\zeta_3}{4} - \frac{\ln^3 2}{6} + \frac{\pi^2 \ln 2}{12}$
$[1, 2, 1/2, 1]$	$-\zeta_3 + \frac{\pi^2 \ln 2}{4}$
$[2, 1, -1, -1]$	$\frac{\zeta_3}{8}$
$[2, 1, -1, 1]$	$-\frac{13\zeta_3}{8} + \frac{\pi^2 \ln 2}{4}$
$[2, 1, 1, -1]$	$-\frac{\pi^2 \ln 2}{4} + \zeta_3$
$[2, 1, 1, 1/2]$	$-\frac{5\zeta_3}{8} + \frac{\pi^2 \ln 2}{6}$
$[2, 1, 1/2, 1/2]$	$-\frac{\ln^3 2}{6} + \frac{\zeta_3}{8}$
$[2, 1, 1/2, 1]$	$-\frac{\pi^2 \ln 2}{4} + \frac{13\zeta_3}{8}$
$[2, 2, 1, -1]$	$-\frac{71\pi^4}{1440} - \frac{\pi^2 \ln^2 2}{6} + \frac{\ln^4 2}{6} + 4\,\mathrm{Li}_4(\frac{1}{2}) + \frac{7 \ln 2 \zeta_3}{2}$
$[2, 2, 1, 1/2]$	$-\mathrm{Li}_4(\frac{1}{2}) - \frac{\ln 2 \zeta_3}{4} - \frac{\pi^2 \ln^2 2}{24} - \frac{\ln^4 2}{24} + \frac{19\pi^4}{1440}$
$[2, 2, 1/2, 1/2]$	$-\frac{\pi^2 \ln^2 2}{24} + \frac{\ln^4 2}{24} + \frac{\pi^4}{1440} + \frac{\ln 2 \zeta_3}{4}$
$[2, 2, 1/2, 1]$	$-3\,\mathrm{Li}_4(\frac{1}{2}) - \frac{\pi^2 \ln^2 2}{8} - \frac{\ln^4 2}{8} + \frac{7\pi^4}{288}$

ACKNOWLEDGMENTS

I used Claude (Anthropic) to assist with the literature search, with writing and debugging the computer-algebra code of the threshold construction and validation and the analysis and checking scripts, with the figures and with the drafting of the text; I have reviewed and verified all results and text.

Appendix A: Endpoint constants

The boundary values of the expansion contain the constants

$$\mathrm{MB}[p, q, a, w] = \frac{a}{(p-1)!} \int_0^1 \mathrm{d}t \, \frac{[-\ln t]^{p-1}}{1 - at} \, \phi_q(wt), \quad \text{(A1)}$$

with $\phi_1(z) = -\ln(1 - z)$ and $\phi_q(z) = \mathrm{Li}_q(z)$ for $q \geq 2$. Besides the elementary values $\mathrm{MB}[2, 2, 1, 1] = \pi^4/120$ and $\mathrm{MB}[2, 1, 1, 1] = \zeta_3$, thirteen distinct constants occur. They were evaluated by quadrature at high precision, independently of the amplitude data, and reduced to the basis of Eq. (21) with the PSLQ algorithm [36] at 110 digits. Table VII lists the results; the largest residual is 4×10^{-107}, and the same relations were obtained in Mathematica.

Appendix B: Normalisation of the published remainders

Five checks fix the normalisation of the tree, one-loop and two-loop amplitudes, Eq. (18). They use the tree amplitudes and the one- and two-loop remainders released with Ref. [5], and the Mathematica code of the original calculation.

Tree level. The projections of the two tree diagrams from the original calculation give Eq. (9), after removal of a colour sum $N_c^2 - 1$ that the file for T_1 carries and the file for T_2 does not. A direct evaluation of the two diagrams with the vertex $iC_G\varepsilon_{\mu\alpha\rho\beta}k_1^\mu k_2^\rho \delta^{ab}$, which follows from Eq. (1), gives $|\mathcal{A}_{\pm\pm}| = C_G^2|\mathcal{M}_1^{(0)}|$ and $|\mathcal{A}_{\pm\mp}| = C_G^2|\mathcal{M}_2^{(0)}|$ to 14 digits.

Tree and one loop. The coefficient of n_f in $\mathcal{M}_i^{(1)}$, divided by $\mathcal{M}_i^{(0)}$, is the same for $i = 1, 2$ at threshold, and its imaginary part equals $-2\pi/3$ to 15 digits at every tested point. This is the value produced by the subtraction term $2I^{(1)}\mathcal{M}^{(0)}$ of Eq. (17) through the n_f part of β_0 and $\ln(-\hat{s}/\mu_R^2)$; the one-loop n_f diagrams have no $\hat{s}$-channel cut. It fixes the normalisation and phase of the tree amplitude relative to the one-loop remainders.

Scale dependence at one loop. A remainder with the dimension of m_A^2 can depend on m_A and μ_R only through their ratio, apart from the overall factor m_A^2. The published one-loop remainders violate this: when m_A and μ_R are multiplied by a common factor λ at fixed β_A and c, $\mathcal{M}_{i,\text{publ}}^{(1)}/m_A^2$ changes by $-2\beta_0 \ln\lambda^2\, \mathcal{M}_i^{(0)}/m_A^2$, to a relative accuracy of 7×10^{-56}. They are independent of μ_R, since their eight $\ln\mu_R^2$ terms cancel. In the original calculation the amplitude is renormalised with Z_{GG}^2, $Z_{GG} = Z_{a_s}$, and $C_G \propto a_s$; with the subtraction operators of Eq. (17), whose scale dependence enters through $(\mu_R^2/(-\hat{s}))^\epsilon$, renormalisation group invariance of Eq. (16) then requires

$$\frac{\partial \mathcal{M}^{(1)}}{\partial \ln \mu_R^2} = 2\beta_0\, \mathcal{M}^{(0)}. \tag{B1}$$

The first line of Eq. (18) restores this dependence; the corrected remainders are invariant under the rescaling to 6×10^{-57}.

Scale dependence at two loops. With the same ingredients, and with the subtraction operator $I^{(2)}$ of the original code, which, unlike Eq. (3.4) of Ref. [5], does not multiply the constant

$$H_g^{(2)} = \tfrac{1}{2}\Big[C_A^2\big(-\tfrac{5}{12} - \tfrac{11}{24}\zeta_2 - \tfrac{1}{2}\zeta_3\big) + C_A n_f\big(\tfrac{29}{27} + \tfrac{\zeta_2}{12}\big) - \tfrac{1}{2}C_F n_f - \tfrac{5}{27}n_f^2\Big] \tag{B2}$$

by $(-\hat{s}/\mu_R^2)^\epsilon$, one finds

$$\frac{\partial \mathcal{M}^{(2)}}{\partial \ln \mu_R^2} = 3\beta_0\, \mathcal{M}^{(1)} + \big(2\beta_1 - 8H_g^{(2)} - 12C_F n_f\big)\mathcal{M}^{(0)}, \tag{B3}$$

where $\beta_1 = \frac{34}{3}C_A^2 - 2C_F n_f - \frac{10}{3}C_A n_f$ and the last term comes from the counterterm of the O_G–O_J amplitude, Eq. (3), in the renormalisation of Ref. [5]. I fitted $\partial\mathcal{M}^{(2)}/\partial\ln\mu_R^2 = a\,\beta_0\mathcal{M}^{(1)} + b\,\mathcal{M}^{(0)}$ to the series at two velocities and angles, for $(N_c, n_f, m_A) = (3, 0, 300\,\text{GeV})$, $(3, 5, 125\,\text{GeV})$ and $(4, 2, 150\,\text{GeV})$. With both lines of Eq. (18), $a = 3$ to 2×10^{-7} and b agrees with Eq. (B3) to 2×10^{-5}, the accuracy of the series at these velocities; the second derivative equals $6\beta_0^2\mathcal{M}^{(0)}$, the value implied by Eqs. (B1) and (B3), to 1×10^{-8}. Without the division by 64 the fit gives $a = 192$, and without the scale term of the one-loop remainders b differs from Eq. (B3) by $6\beta_0^2\ln(m_A^2/\text{GeV}^2)$; the agreement therefore also fixes the reference scale of the published one-loop remainders at $\mu_R = 1$ GeV.

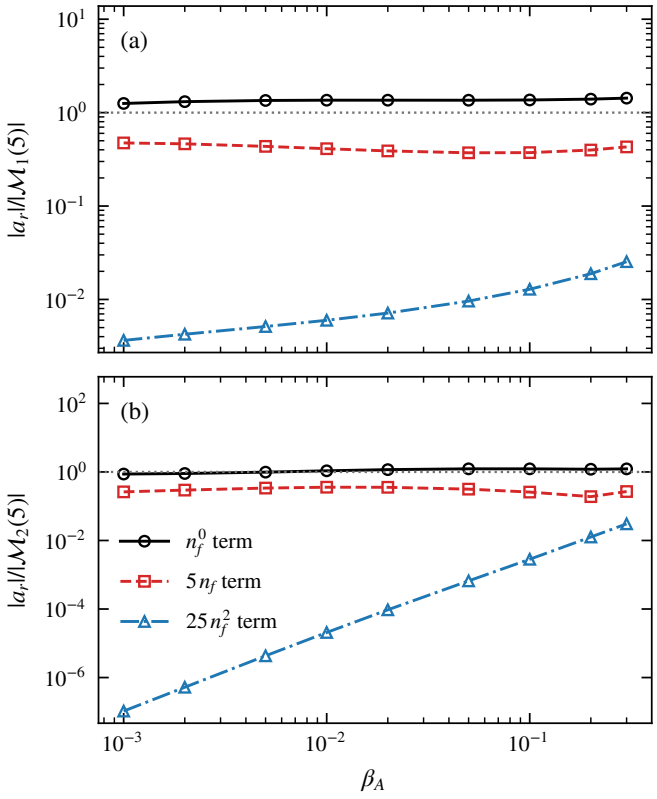


FIG. 6. Magnitudes of the three terms of Eq. (C1), divided by the magnitude of the complete coefficient, for $c = 1/3$, $m_A = \mu_R = 125\,\text{GeV}$ and $N_c = 3$. The horizontal line marks unity.

Origin of the factor 64. The Mathematica code of the original calculation documents it: the two-loop renormalisation step multiplies the O_G–O_G amplitude, including the tree and one-loop amplitudes that enter the infrared subtraction, by $64 = 8^2$, whereas the one-loop step does not. The pole cancellation of Ref. [5] is unaffected, since all terms of the two-loop subtraction carry the same factor. The released code and data use the normalisation of Eq. (18).

The original output. The Mathematica output of the original calculation confirms both corrections. Its one-loop remainders, normalised to the tree amplitude, depend on μ_R as in Eq. (B1) and agree with the first line of Eq. (18) to 5×10^{-56}; the published one-loop files coincide with them at $\mu_R = 1$ GeV. An earlier version of the two-loop renormalisation step, without the factor 64 and with the O_J contribution switched off, was evaluated at $x = 1/11$, $z = 3/11$. At that point the published P_1 remainder equals 64 times this result, multiplied by the tree amplitude, plus the O_G–O_J counterterm of the published version; the real parts agree to 5×10^{-31}. The imaginary parts differ by $32C_F$

TABLE VIII. Evaluation time for the two complete Type II coefficients at the benchmark parameters, $c = 0$ and 80-digit working precision, for the unexpanded remainders (exact) and the series. First calls start with empty caches; repeated times are medians of seven evaluations at the same point. Starting the kernel and loading the files are excluded.

	first call [s]			repeated [s]		
β_A	exact	series	ratio	exact	series	ratio
1/10	56.04	11.61	4.83	0.3422	0.0178	19.18
1/5	45.68	11.64	3.92	0.3403	0.0179	18.99
3/10	44.64	11.64	3.84	0.3485	0.0181	19.22

times the one-loop O_G–O_J amplitude, a multiple of the tree amplitude. That earlier run is not an alternative result, however. Its renormalisation constant exceeds $Z_{GG}^2 = Z_{a_s}^2$ by $2\beta_1 a_s^2/\epsilon$, a pole with C_A^2 terms that no O_J contribution can supply, and it contains no O_G–O_J counterterm. With the same infrared subtraction as the published version, its single poles cannot cancel, and the difference is not an uncertainty of the published remainder. In the published version the normalisation of the O_G–O_J counterterm was fixed by the cancellation of the $C_F n_f$ single pole; a record of the original calculation from June 2019 shows that the same factor is required in both projections at seven kinematic points. The files and a script that reproduce these comparisons are part of the released material.

Appendix C: Flavour decomposition

For $n_f = 5$ the complete coefficients are the sums

$$\begin{aligned} \mathcal{M}_i^{(2)}(5) &= \mathcal{M}_{i,0}^{(2)} + 5\,\mathcal{M}_{i,1}^{(2)} + 5^2\,\mathcal{M}_{i,2}^{(2)} \\ &\equiv a_0 + a_1 + a_2. \end{aligned} \tag{C1}$$

Figure 6 shows $|a_r|/|\mathcal{M}_i^{(2)}(5)|$. The n_f^0 term dominates both coefficients. The n_f^1 term is sizeable and partly cancels it: for $\mathcal{M}_1$ the quantity $1-|a_0+a_1+a_2|/(|a_0|+|a_1|+|a_2|)$ lies between 42.2% and 46.9% over the velocities shown. At leading power the number of flavours enters with the sign opposite to the gluonic terms, through $\beta_0 = (11N_c - 2n_f)/3$ in $\widehat{A}_{12}$, through $N_c a_{\mathrm{A}} - n_f a_{\mathrm{F}}$ in $\widehat{A}_{11}$ and through $4N_c - n_f$ in $\widehat{A}_{21}$ [Eqs. (28) and (29)], which accounts for the cancellation at small β_A. The n_f^2 term of $\mathcal{M}_2$ is suppressed as $\beta_A^2(1-c^2)$, Eq. (24).

Appendix D: Evaluation time and released data

The series through β_A^6 occupy 40 kB, against 204 MB for the published two-loop files, a factor of 5061. The evaluation time was measured before the reduction of Sec. III B, with the unexpanded expressions and the series in the forms used for the comparison of Sec. III C, at $\beta_A = 1/10$, 1/5 and 3/10, $c = 0$, $m_A = \mu_R = 125\,\mathrm{GeV}$ and $N_c = 3$, with 80-digit working precision. On a first evaluation the series is 3.84–4.83 times faster. For repeated evaluations at the same point it is 18.99–19.22 times faster, with median 19.18; this does not correspond to a scan over new points. The two timed representations occupy 0.61 MiB and 42.77 MiB on disk, a factor of 70.4. Table VIII gives the individual measurements. The measurements used Wolfram Language 13.2.0 for Linux x86 under Ubuntu 24.04.2 LTS on an AMD Ryzen 9 7950X processor with 124.9 GiB of memory.

The Supplemental Material [24], which is also attached to the arXiv version of this article as ancillary files, is a self-contained package. It contains the construction files, one for each flavour coefficient and power of β_A from β_A^{-4} to β_A^6; the series through β_A^6 in Wolfram Language form; the reductions of the endpoint constants; the 1080 validation records and the forward angle data; the tree-level projections; and an evaluator in Python and Wolfram Language. It also contains the Wolfram Language, C++ and Python code of the threshold construction and of the validation, the outputs of the original runs, against which the construction files and the validation records are checked, and a script that repeats the construction from the published expressions. The published expressions of Ref. [5] are downloaded and checked against checksums by a script. Each table, figure and number in this paper is generated from these data by a single command, including the check of the DPS weight in Fig. 3, the logarithms of Sec. V and the normalisation checks of App. B. A test suite checks the evaluator against the validation records. Only the construction of the series from the unexpanded expressions requires Mathematica. The series are intended for $\beta_A \le 0.2$ at the angles tested (Sec. V C); outside this range the unexpanded remainders should be used.

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